Digital Twin of a Rotating Shaft for Predictive Maintenance: Remaining Useful Life Prediction using MATLAB Simulink and Simscape

Intelligent Systems Case Study

By

Group- 9

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Contents

Abstract

Rotating shafts are critical components in industrial machinery, and unexpected failures can lead to costly downtime and secondary damage. This project presents a digital twin of a rotating shaft developed in MATLAB Simulink and Simscape to support predictive maintenance through remaining useful life (RUL) estimation. The physics-based model generates sensor-like signals, including vibration, torque, and angular speed, which are processed into time- and frequency-domain features. Progressive degradation is simulated through controlled fault-severity injection to produce lifecycle data and a corresponding health degradation trend. A Bidirectional LSTM with an attention mechanism is trained on 30-day feature sequences to predict RUL in days. Results are visualized using a MATLAB App Designer dashboard that displays the health index, degradation severity, and predicted RUL, and provides user controls and warning messages. The proposed workflow demonstrates an end-to-end framework that combines physics-based simulation and data-driven prognostics and can be extended for calibration and validation using real machine data.

Keywords: Digital Twin, Rotating Shaft, Simulink, Simscape, Prognostics, Health Index, Remaining Useful Life, BiLSTM, Attention.

1 Introduction

Rotating shafts are widely used in industrial machines such as pumps, compressors, turbines, gearboxes, and drive trains. Because these components operate continuously and often under variable load conditions, they are exposed to progressive degradation that can ultimately lead to failure. Condition changes typically appear first in measurable signals such as vibration, torque, and speed, making rotating shafts a common target for predictive maintenance and prognostics. This project investigates an end-to-end approach for remaining useful life (RUL) prediction by combining a physics-based digital twin with data-driven machine learning.

1.1 Background

Unexpected shaft-related failures can lead to significant downtime and can also cause secondary damage to couplings, bearings, and motors. Traditional maintenance strategies are often insufficient for such systems. Corrective maintenance reacts only after failure occurs, while preventive maintenance schedules service at fixed intervals, which can result in premature replacement or delayed intervention. Predictive maintenance aims to address these limitations by using condition information to estimate health and forecast failure, allowing maintenance to be scheduled based on actual system state [1, 2].

1.2 Motivation

Digital twins support predictive maintenance by linking a physics-based representation of the asset with analytics models that interpret sensor data. A physics-based twin is useful not only for monitoring but also for generating controlled datasets, testing fault scenarios, and understanding how degradation affects measurable outputs. This is particularly relevant when real degradation datasets are limited, expensive to obtain, or difficult to label with ground-truth RUL. Deep learning methods have been shown to be particularly effective for learning degradation patterns directly from condition monitoring data, outperforming traditional feature-based approaches in many machine health monitoring applications [3].

MATLAB Simulink and Simscape provide an integrated environment for developing such a workflow. They enable dynamic system modeling, sensor signal logging, and direct integration with signal processing and machine learning pipelines, which supports rapid iteration from a mechanical model to an RUL predictor.

1.3 Problem Statement

The aim of this project is to develop an end-to-end framework for RUL estimation of a rotating shaft system. The framework should (i) generate sensor-like signals from a physics-based digital twin, (ii) extract features that capture degradation progression under progressive fault injection, and (iii) predict remaining useful life using a Bidirectional LSTM model with an attention mechanism. In addition, the results should be presented in an interpretable way to support monitoring and maintenance decisions.

1.4 Objectives

The project objectives are to build a rotating shaft digital twin in MATLAB Simulink and Simscape, generate vibration/torque/speed signals, and construct feature sequences that represent system health over time. Using simulated lifecycle data created by increasing fault severity, a health degradation trend is obtained and used to train a Bidirectional LSTM with attention for RUL prediction. Finally, a monitoring dashboard is implemented to visualize the health index, degradation severity, and predicted RUL, and to provide warnings and operational controls.

1.5 Scope and Assumptions

This work is conducted in a simulation-driven setting. Degradation is introduced through controlled parameter changes that represent increasing fault severity. The RUL model is trained and evaluated using synthetic lifecycle sequences generated from the digital twin, and the results therefore reflect the assumptions and fidelity of the simulation. Calibration using measured data from a physical system and validation across multiple operating regimes are identified as future improvements.

1.6 Report Organization

The remainder of this report is organized as follows. Section 2 describes the tools used and the overall methodology. Sections 3 and 4 present the digital twin model and the signal/feature pipeline. Section 5 details the RUL prediction model based on a Bidirectional LSTM with attention. Sections 6 and 7 present results and validation, followed by discussion in Section 8. Section 9 concludes the report and outlines future work.

2 Materials and Methods

This section describes the tools, software environment, and methodology used to develop the rotating shaft digital twin and the remaining useful life prediction workflow. The overall approach follows a pipeline that begins with physics-based simulation, continues with signal acquisition and feature extraction, and concludes with a data-driven prognostics model and a monitoring dashboard.

2.1 Software Tools and Technology Stack

The implementation was carried out using MATLAB as the primary environment for modeling, simulation, analytics, and visualization. Simulink was used for time-domain simulation and system integration, while Simscape was used to represent the physical rotating shaft dynamics and mechanical interactions. Signal Processing Toolbox functions were applied to compute time- and frequency-domain indicators from the simulated sensor outputs. The prognostics model was developed using deep learning functionality in MATLAB, and the final visualization and monitoring interface was implemented using MATLAB App Designer. Version control and collaboration were managed using GitHub.

2.2 Overall Methodology

The workflow begins with defining operating conditions and model parameters, followed by building the rotating shaft digital twin in Simulink/Simscape and instrumenting it with virtual sensors. Simulation runs are performed to generate vibration, torque, and speed signals. To obtain lifecycle data for prognostics, degradation is simulated through progressive fault injection, producing sequences that represent aging behavior.

From the logged signals, a feature vector is extracted over fixed time windows. These features are normalized using a healthy baseline to form an interpretable health representation and to build a degradation trend over time. The feature sequences are then used to train a Bidirectional LSTM model with attention, which predicts the remaining useful life of the system. Finally, the results are integrated into a dashboard that displays the health index, severity progression, and predicted RUL. The overall workflow follows established digital twin–driven predictive maintenance pipelines reported in the literature, where physics-based simulation is combined with data-driven prognostics models for remaining useful life estimation [1, 2].

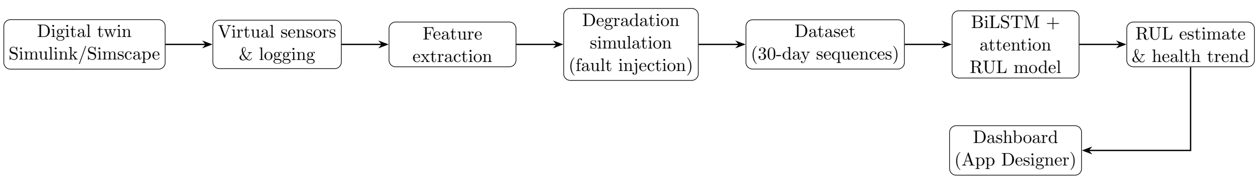

Figure 1: Overall workflow from physics-based digital twin simulation to feature extraction, RUL prediction, and dashboard visualization.

2.3 System Architecture

The system can be viewed as four connected layers. The physical layer consists of the Simscape-based rotating shaft model and its excitation. The sensing layer provides virtual measurements for vibration, torque, and angular speed. The analytics layer includes feature extraction and the BiLSTM-attention model for RUL prediction. The visualization layer presents the outputs through a dashboard, enabling interpretation and user interaction.

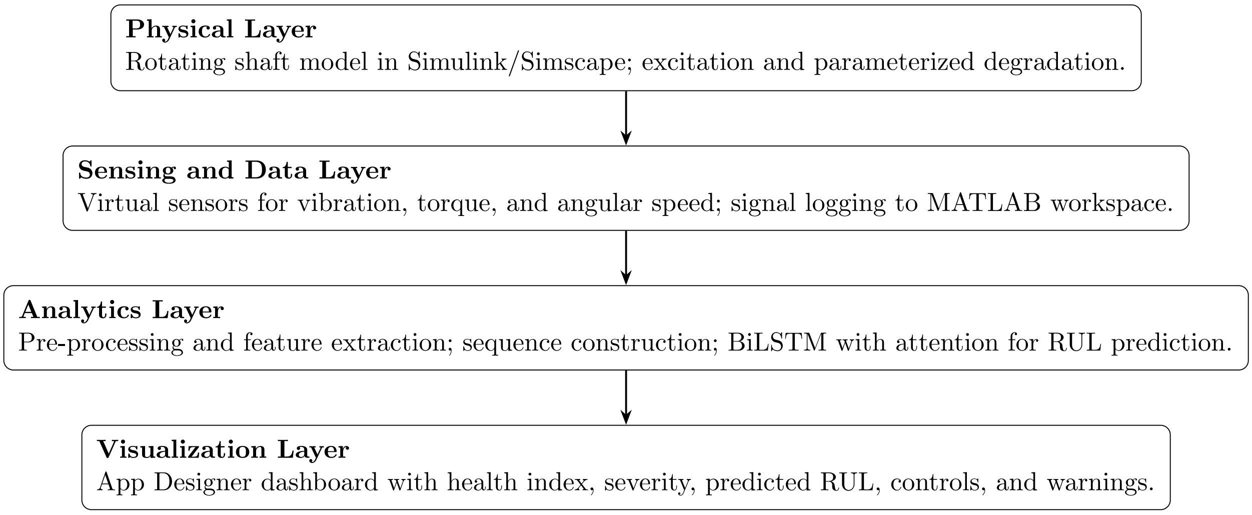

Figure 2: System architecture of the rotating shaft digital twin and prognostics pipeline.

2.4 Degradation Simulation and Dataset Generation

Since real run-to-failure datasets were not available for this project, degradation trajectories were generated in simulation. Fault severity was increased progressively to emulate realistic wear evolution and to produce health degradation over operating time. The resulting lifecycle data were organized into fixed-length sequences for machine learning, with each sequence representing a 30-day history of extracted features and the corresponding RUL label in days.

2.5 Feature Extraction Strategy

To create inputs suitable for prognostics, time-series signals were segmented into fixed-length windows. For each window, statistical and spectral features were computed to capture the energy, impulsiveness, and frequency content of the vibration response. The extracted feature vectors were then assembled into sequences and normalized to reduce scale differences and improve model training stability.

2.6 RUL Prediction Model

RUL estimation was performed using a Bidirectional LSTM network with an attention mechanism. The network takes a sequence of daily feature vectors as input and outputs a single RUL estimate. Model training, evaluation, and tuning were carried out using standard regression metrics and validation procedures, which are described in later sections together with the experimental results.

2.7 Dashboard and Monitoring

A MATLAB App Designer dashboard was developed to present the outputs of the pipeline. The dashboard provides plots of the health index, degradation severity, and predicted RUL, along with controls to start, pause, and stop updates. A status panel displays the current RUL estimate and warning messages when degradation accelerates or thresholds are approached.

3 Digital Twin Modeling

This section describes the physics-based digital twin of the rotating shaft developed in MATLAB Simulink and Simscape. The model is designed to reproduce the essential rotational dynamics of the shaft system and to generate sensor-like signals that are later used for feature extraction, degradation modeling, and remaining useful life (RUL) prediction.

3.1 Model Overview

The digital twin is implemented as an integrated Simulink–Simscape model of a rotational mechanical system. The model accepts an excitation input representing the commanded actuation and produces measurable responses in terms of angular speed, shaft torque, and a vibration-related output signal. The structure is parameterized to support repeatable simulation experiments and controlled degradation injection for lifecycle dataset generation.

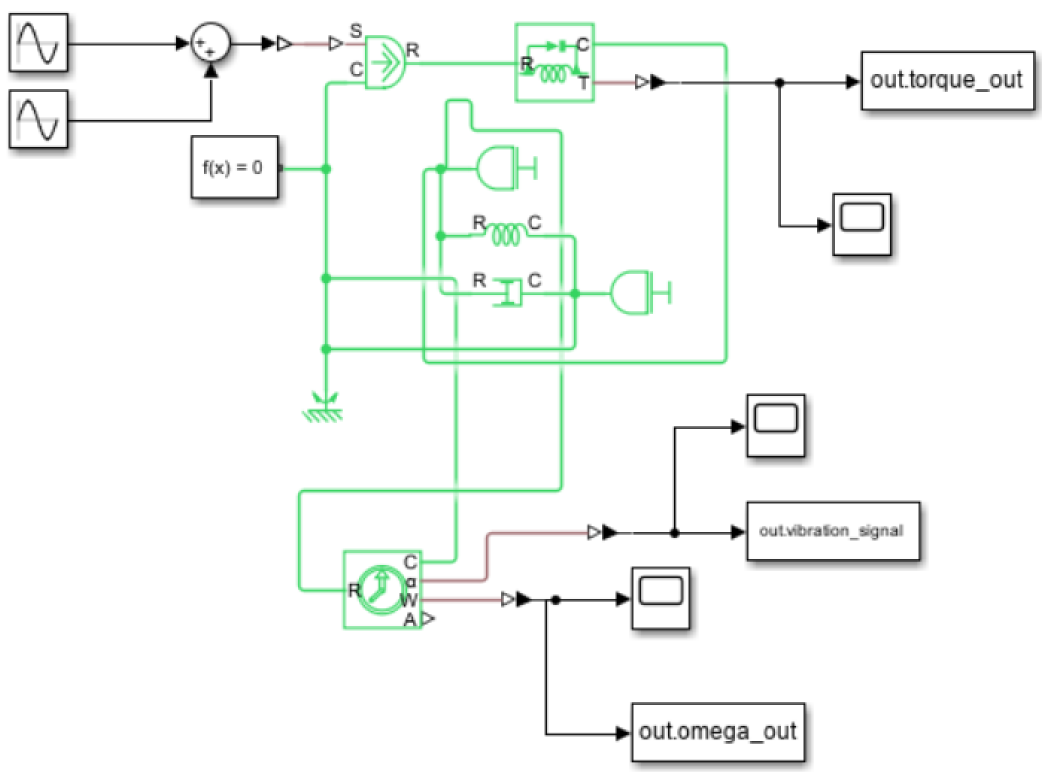

Figure 3: Simulink/Simscape implementation of the rotating shaft digital twin with virtual sensing and signal logging.

3.2 Rotational Dynamics Model

The core mechanical behavior is captured using standard rotational elements. The shaft inertia represents energy storage and governs the transient response to applied torque, while viscous damping captures frictional and dissipative losses. A mechanical reference is included to define a consistent physical ground, ensuring that the torque and angular velocity states are well-posed and numerically stable during simulation.

3.3 Excitation and Input Integration

The excitation signal is formed by combining the commanded input with optional disturbance terms to represent operating variability. A gain stage is used to convert the input signal into a physical actuation quantity, such as applied torque. This torque drives the rotational network, resulting in speed and torque responses consistent with the modeled dynamics.

3.4 Virtual Sensing and Signal Logging

To enable analytics and prognostics development, the model is instrumented with virtual sensors that measure key outputs. Angular speed and shaft torque are obtained directly from the rotational domain, while the vibration-related signal is derived from the mechanical response of the system. All outputs are logged at a fixed sampling rate and exported to the MATLAB workspace. This configuration supports consistent post-processing and ensures that feature extraction can be repeated across multiple degradation scenarios.

3.5 Degradation Parameterization

Degradation is introduced in a controlled manner by progressively increasing fault severity through parameter changes within the digital twin. This approach generates time histories that exhibit health deterioration over operating time, enabling the creation of synthetic lifecycles for RUL learning. The degradation mechanism is configurable so that different degradation rates and patterns can be generated to improve the diversity of the training dataset.

3.6 Simulation Configuration

All experiments are executed using consistent solver settings and logging configurations to ensure comparability across runs. The selected step size provides sufficient time resolution for both time-domain statistics and frequency-domain analysis used during feature extraction. The resulting simulation outputs form the basis for subsequent health indexing and RUL prediction.

4 Data Pipeline and Feature Engineering

This section describes how simulation outputs from the digital twin are converted into structured datasets suitable for degradation tracking and remaining useful life prediction. The pipeline includes signal logging, segmentation into fixed horizons, feature extraction in time and frequency domains, and construction of 30-day sequences used as inputs to the BiLSTM-attention model.

4.1 Signal Acquisition and Logging

The digital twin generates sensor-like time-series outputs for vibration, shaft torque, and angular speed. These signals are logged at a fixed sampling interval and exported to the MATLAB workspace to ensure consistent post-processing. To support repeatable dataset generation, logging variables, sample rates, and simulation duration are kept consistent across runs.

2531	307	12217
2877	-77	15060
2415	-414	12012

Figure 4: Example time-domain signals logged from the digital twin.

4.2 Segmentation and Sequence Construction

For prognostics, signals are organized into a sequence format representing the system history. The continuous sensor signals are segmented into fixed windows, and a feature vector is computed for each window. Feature vectors are then aggregated into a fixed-length history representing 30 days of operation, with each day represented by a 12-dimensional feature vector. This produces an input tensor of size (30×12) per lifecycle segment, matching the input requirements of the RUL prediction model.

4.3 Feature Extraction

Feature extraction is performed to summarize the signal content in a compact and informative representation. Both time-domain and frequency-domain indicators are used to capture changes in signal energy, impulsiveness, and spectral distribution associated with degradation progression. The selected features are designed to be computationally efficient and stable under measurement noise.

Table 1: Feature set used to represent daily condition for prognostics.

Feature	Domain	Interpretation
RMS vibration	Time	Overall vibration energy level
Peak amplitude	Time	Maximum observed amplitude
Peak-to-peak amplitude	Time	Signal range within the window
Crest factor	Time	Peak-to-RMS ratio (impulsiveness)
Kurtosis	Time	Distribution tail heaviness (impacts/impulses)
Skewness	Time	Distribution asymmetry
Dominant frequency	Frequency	Frequency with maximum magnitude
Low-band energy	Frequency	Spectral energy in a low-frequency band
Mid-band energy	Frequency	Spectral energy in a mid-frequency band
High-band energy	Frequency	Spectral energy in a high-frequency band
Mean speed (RPM)	Operating	Average rotational speed
Mean voltage / input	Operating	Input level corresponding to excitation

4.4 Normalization and Baseline Referencing

To improve model training stability and enable comparison across operating histories, features are normalized relative to a healthy baseline and scaled to a common range. Baseline referencing reduces the influence of absolute signal magnitude and emphasizes deviation from nominal behavior, which is more directly related to degradation progression.

4.5 Health Index Construction

A health index is constructed from the feature vectors to provide an interpretable degradation indicator. The health index is designed to decrease as fault severity increases, producing a trend suitable for monitoring and for learning degradation dynamics. The same health trend is also displayed in the monitoring dashboard to support interpretation of the predicted RUL.

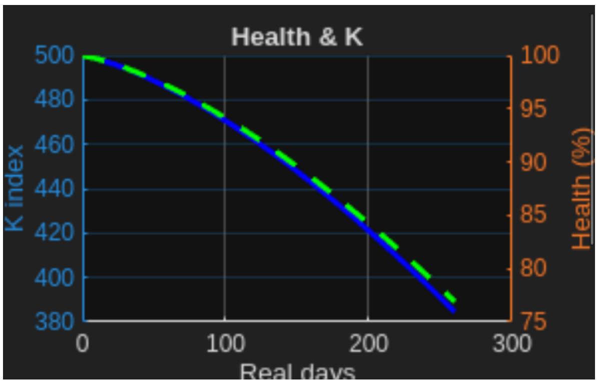

Figure 5: Example health index trend generated under progressive degradation injection.

5 RUL Prediction Model

This section describes the data-driven prognostics model used to estimate remaining useful life (RUL) from condition histories generated by the rotating shaft digital twin. The objective is to learn a mapping from recent condition evolution, represented as a fixed-length sequence of engineered features, to a scalar RUL estimate in days. A Bidirectional Long Short-Term Memory (BiLSTM) network is adopted to capture temporal dependencies in the degradation trajectory, and an attention mechanism is applied to improve both predictive performance and interpretability by weighting the most informative time steps. Recurrent neural network architectures, including LSTM-based models, are widely used for remaining useful life prediction due to their ability to capture temporal degradation dependencies [3].

5.1 Problem Formulation

RUL estimation is formulated as a supervised sequence-to-value regression task. For each sample, the input is a condition history of length $T = 30$ days, where each day is represented by a feature vector $\mathbf{x}_t \in \mathbb{R}^{12}$. The model receives the sequence $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_{30}] \in \mathbb{R}^{30\times 12}$ and outputs a single continuous value $\hat{y} \in \mathbb{R}$ representing the predicted remaining useful life in days.

5.2 Dataset Generation and Labeling

Since run-to-failure measurements were not available for this study, lifecycle datasets were generated using the digital twin by progressively injecting degradation through a controlled fault-severity profile. For each simulated lifecycle, daily feature vectors were extracted and assembled into fixed-length sequences. Ground-truth RUL labels were derived from the remaining time to reach a predefined end-of-life threshold on the degradation trajectory. This controlled setup provides consistent supervision for training while allowing generation of multiple degradation patterns and rates.

5.3 Pre-processing and Normalization

All features are normalized prior to model training to reduce scale differences and improve numerical stability. To better reflect realistic operating conditions, controlled noise is introduced into the feature sequences to emulate measurement variability. Where long histories contain missing values, interpolation is applied to maintain consistent sequence length and to prevent discontinuities that can degrade recurrent model training.

5.4 Model Architecture

The prediction model is based on a BiLSTM backbone followed by an attention mechanism and a regression head. The BiLSTM processes the input sequence in both forward and backward directions, enabling the network to represent degradation behavior using context from the full history window. The attention layer assigns weights to individual time steps, allowing the model to emphasize periods that contribute most strongly to the RUL estimate, such as intervals where degradation accelerates. A fully connected regression head maps the attention-weighted representation to the final scalar RUL prediction.

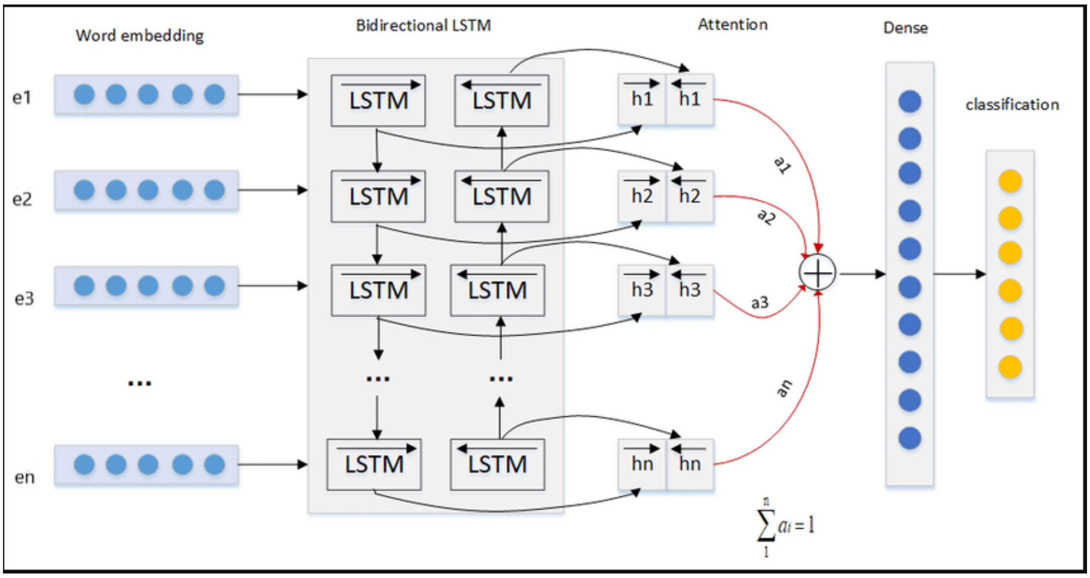

Figure 6: Architecture of the BiLSTM-attention model used for RUL prediction from 30-day feature sequences.

5.5 Training Configuration

Training is performed using mini-batch optimization with a validation split for model selection. Regularization is applied to reduce overfitting and improve generalization across degradation profiles, including dropout within the recurrent network. Early stopping is used to terminate training when validation performance no longer improves, and the best-performing model parameters are retained. Gradient stabilization techniques are applied to prevent exploding gradients during backpropagation through time.

5.6 Performance Metrics

Model performance is evaluated using standard regression metrics with direct interpretation in days. Mean Absolute Error (MAE) is reported as the primary metric due to its interpretability, while Root Mean Square Error (RMSE) is used to emphasize large prediction deviations. The coefficient of determination (R^2) is used to quantify explained variance. In addition, tolerance-based accuracy is reported as the fraction of predictions within a specified error margin (e.g., within ± 10 days), reflecting practical usefulness in maintenance planning.

5.7 Deployment within the Prognostics Pipeline

After training, the RUL model is integrated into the digital-twin workflow. As new sensor data is generated and converted into feature vectors, the latest 30-day window is formed and passed to the trained model to obtain an updated RUL estimate. These estimates are visualized in the monitoring dashboard together with the health index and degradation severity, supporting human interpretation of current condition and projected remaining life.

6 Results

This section presents the main outcomes of the project, focusing on the end-to-end operation of the proposed workflow. The results include signal generation from the rotating shaft digital twin, construction of degradation histories through progressive fault-severity injection, RUL predictions produced by the BiLSTM-attention model, and visualization of model outputs in the monitoring dashboard.

6.1 Simulation Output and Logged Signals

The digital twin was executed under consistent operating conditions and solver settings to generate sensor-like time series. The primary logged outputs were vibration, shaft torque, and angular speed. These signals form the basis for the downstream analytics pipeline, where features are computed and assembled into fixed-length histories for RUL estimation.

2531	307	12217
2877	-77	15060
2415	-414	12012

Figure 7: Example time-domain signals generated by the digital twin (vibration, torque, and angular speed).

6.2 Degradation Simulation and Lifecycle Generation

To obtain lifecycle data for prognostics, degradation was introduced through a progressive fault-severity injection mechanism. Increasing severity produced measurable changes in the simulated sensor signals and therefore in the extracted condition representation over time. This process enabled the generation of repeatable degradation trajectories used to construct 30-day sequences for training and evaluation of the RUL model.

6.3 RUL Prediction Results

The BiLSTM-attention model predicts remaining useful life in days based on the most recent 30-day history of condition features. The predicted RUL decreases as degradation progresses, and the prediction can be updated sequentially as new feature windows become available. To evaluate prediction quality, the model outputs are compared against ground-truth RUL labels derived from the simulated end-of-life threshold crossing.

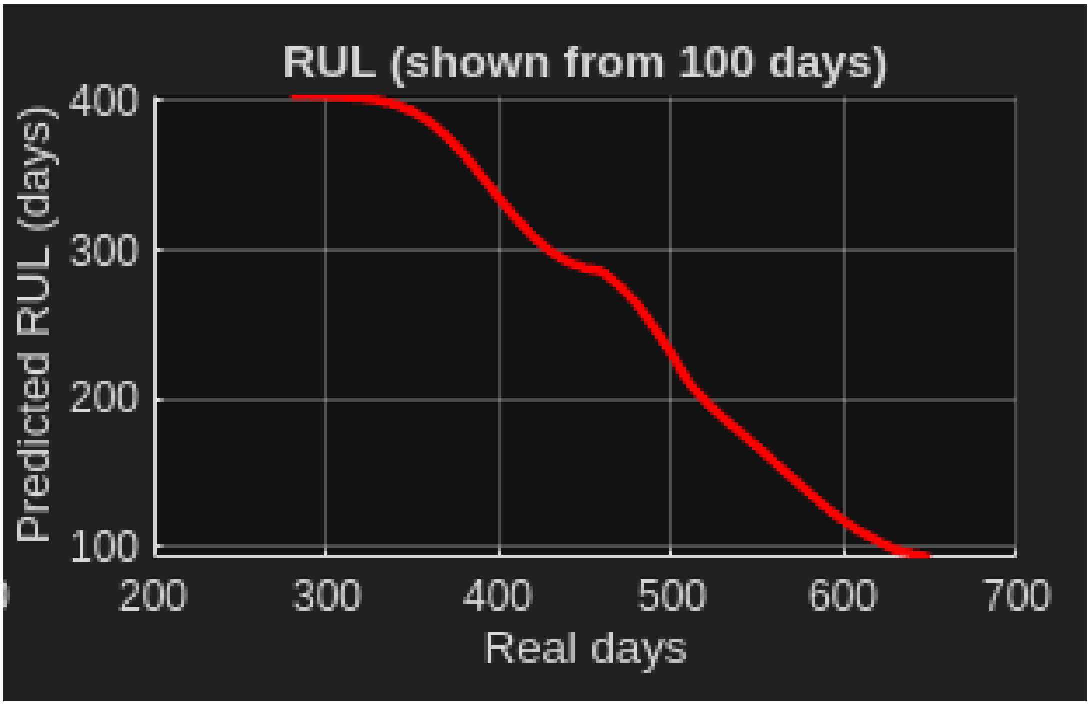

Figure 8: Predicted versus ground-truth RUL for the evaluation set.

6.4 Performance Summary

Performance is summarized using regression metrics with direct interpretation in days. Mean Absolute Error (MAE) captures average prediction error, while RMSE penalizes larger errors. In addition, tolerance-based accuracy can be reported to reflect practical maintenance usefulness (e.g., percentage of predictions within ±10 days).

Table 2: RUL prediction performance on the evaluation set.

Metric	Value	Unit	Notes
MAE	*(fill)*	days	Average absolute error
RMSE	*(fill)*	days	Penalizes large errors
R^2	*(fill)*	–	Variance explained
Accuracy (±10 days)	*(fill)*	%	Practical tolerance

6.5 Dashboard Visualization

The final outputs of the workflow are presented in a MATLAB App Designer dashboard. The interface consolidates the key prognostics signals by displaying degradation severity, the health indicator trend, and the predicted RUL over time. In addition to visualization, the dashboard provides start/pause/stop controls and a status panel that reports the current RUL estimate and warning messages when the system approaches critical thresholds. This demonstrates how the digital twin and the learned RUL predictor can be used together for continuous monitoring and maintenance decision support.

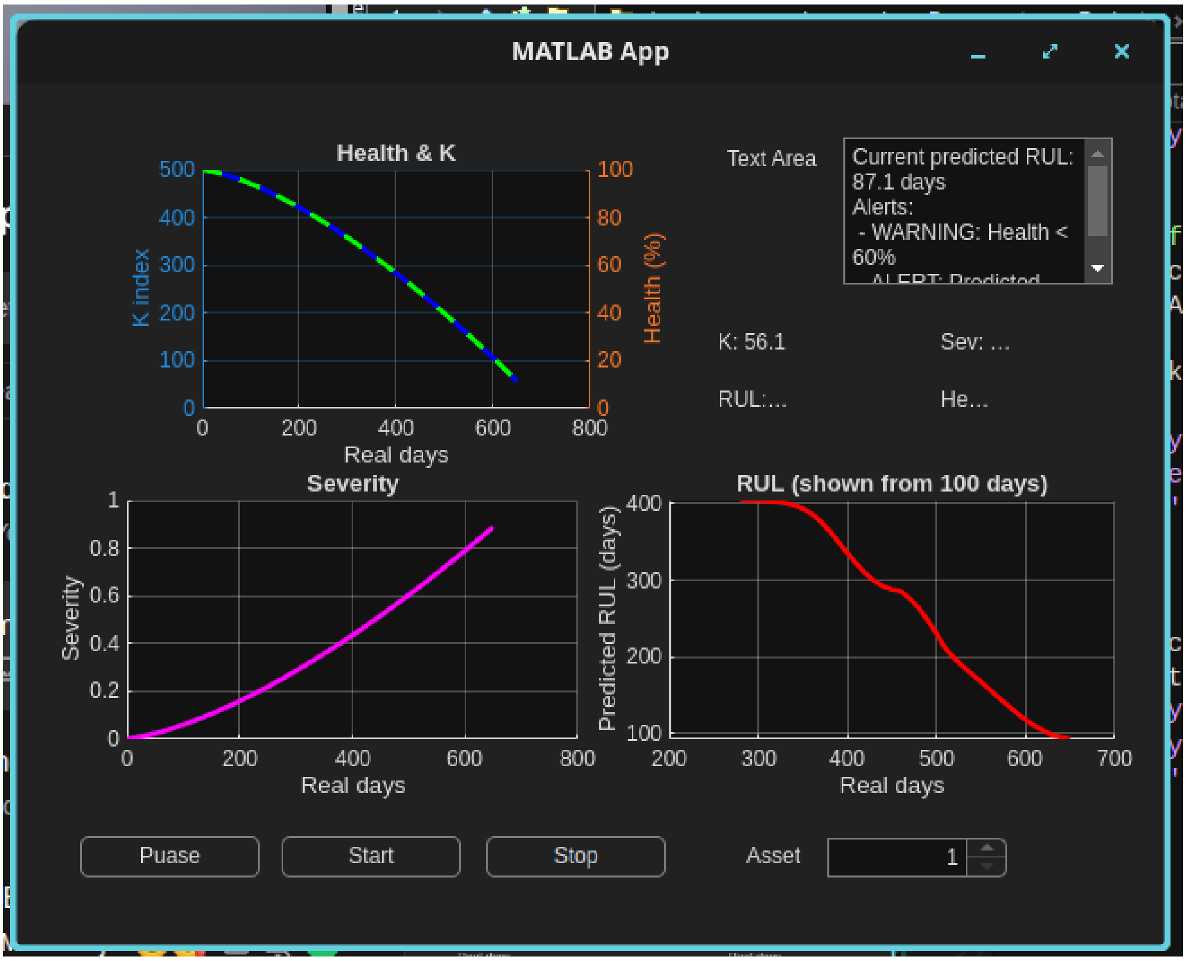

Figure 9: Monitoring dashboard displaying health indicator, degradation severity, and predicted RUL with operational controls and warning messages.

7 Validation

This section describes the validation strategy applied to the digital twin and the RUL prediction workflow. Because the project is simulation-driven and does not include a full run-to-failure experimental campaign, validation is performed by verifying internal consistency of the physics model, confirming that the degradation injection mechanism produces coherent lifecycle trajectories, and evaluating the BiLSTM-attention predictor against ground-truth labels derived from the simulated end-of-life definition. The goal of this section is not to claim industrial certification, but to demonstrate that the proposed pipeline behaves as expected and produces defensible results within the defined scope.

7.1 Digital Twin Verification

The digital twin was first verified for numerical stability and repeatability under nominal operating conditions. The model was executed multiple times using identical inputs to confirm that the logged vibration, torque, and speed outputs remain consistent. Solver settings and logging rates were also verified to ensure that the time resolution is sufficient for the intended feature extraction and that no numerical artifacts dominate the measured signals. These checks establish confidence that downstream prognostics results are driven by modeled dynamics rather than simulation instability.

7.2 Verification of Sensing and Data Logging

Virtual sensors and logging blocks were verified by checking signal ranges, units, and synchronization across outputs. In particular, the vibration-related output was inspected to confirm that it responds to operating changes and degradation injection in a physically interpretable manner. Logging consistency was validated by ensuring that dataset generation produces sequences with the expected length and feature dimensionality, which is critical for training and deploying the BiLSTM-attention model.

7.3 Degradation Injection Consistency

Degradation was introduced through progressive fault-severity injection to emulate aging. Validation focused on confirming that severity evolves according to the intended profile and that increasing severity produces systematic changes in the generated signals. A key requirement for RUL learning is the existence of a coherent degradation trajectory; therefore, the degradation process was checked for monotonic progression and for realistic variability, where short-term fluctuations may occur due to noise but long-term trends remain consistent with worsening condition.

7.4 End-of-Life Definition and Label Consistency

RUL labels were derived from the simulated lifecycle based on a predefined end-of-life (EOL) threshold. The label-generation procedure was validated by confirming that, within each lifecycle, the computed RUL decreases over time and reaches approximately zero at the EOL crossing. This step ensures that supervised learning targets are consistent and that the model is trained to predict a well-defined quantity. In addition, the selected threshold was reviewed to ensure it reflects a meaningful degradation level within the simulation context, rather than an arbitrary numerical value.

7.5 RUL Predictor Evaluation Protocol

The BiLSTM-attention model was evaluated using data not used during training, following a time-based separation between training and testing to reduce leakage of future information. Performance was assessed using standard regression metrics reported in the Results section (MAE, RMSE, and R^2), together with tolerance-based accuracy to reflect practical usefulness in maintenance planning. Beyond numerical metrics, qualitative checks were performed to ensure prediction behavior is physically plausible: predicted RUL should generally decrease as degradation progresses, and updates should remain stable rather than oscillating excessively from one step to the next.

7.6 Robustness and Sensitivity Checks

To assess robustness, the workflow was tested under variations in degradation rate and noise levels used during dataset generation. These checks evaluate how sensitive the RUL estimate is to changes in the degradation trajectory and measurement variability. The dashboard was also used as an integration validation tool by confirming that health, severity, and RUL curves update correctly during runtime and that warning messages trigger consistently as thresholds are approached.

7.7 Limitations of Validation

The primary limitation of the validation is that it is based on simulated degradation rather than measured run-to-failure data. Consequently, prediction accuracy reflects consistency with the digital twin assumptions and the selected degradation mechanism. Real systems introduce additional sources of variability, including sensor mounting effects, temperature dependence, machine-to-machine differences, and operating regime changes. Calibration of model parameters and thresholds using physical data, followed by validation on independent operational datasets, is required before deployment in an industrial environment.

8 Discussion

This section interprets the results of the proposed digital twin and prognostics workflow with a focus on what the outputs imply for predictive maintenance. The discussion emphasizes the relationship between injected degradation, observed signal/feature behavior, the resulting health representation, and the stability of the remaining useful life (RUL) estimates. Key limitations and practical deployment considerations are also addressed.

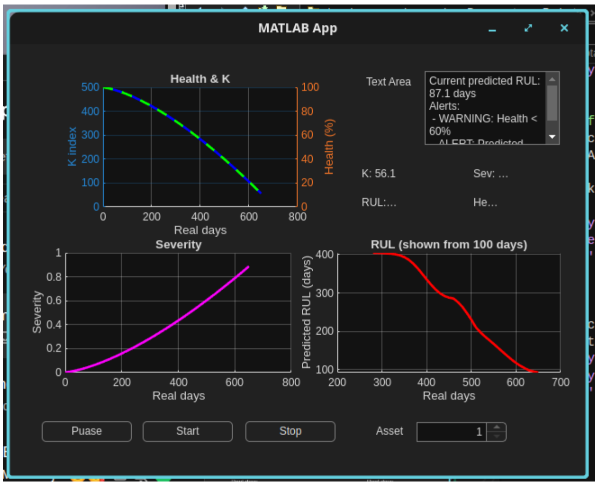

Figure 10: Combined view of degradation severity, health indicator, and predicted RUL used for interpretation and decision support.

8.1 Interpretation of Degradation Behavior

A central outcome of the project is that progressive fault-severity injection produces coherent lifecycle trajectories suitable for prognostics. As severity increases, the simulated sensor signals exhibit systematic changes that propagate through the feature extraction pipeline. This controlled behavior is important because the RUL predictor relies on consistent temporal trends rather than isolated fault signatures. The degradation mechanism therefore acts as the bridge between the physics-based model and the data-driven RUL estimator, enabling supervised learning using synthetic ground truth.

8.2 Health Representation and Monitoring Implications

Normalizing condition features against a healthy baseline provides an interpretable view of system state and simplifies monitoring across lifecycles. The health indicator displayed in the dashboard supports quick assessment of whether the system is trending toward end-of-life. In a practical maintenance context, the health trend can be used to trigger inspections or to prioritize assets even before the RUL estimate reaches critical levels. The combined visualization of severity, health, and RUL also improves traceability by linking the cause (degradation injection), the observed effect (health decline), and the decision variable (RUL).

8.3 RUL Prediction Behavior and Practical Meaning

The BiLSTM-attention model uses a 30-day history of features to estimate RUL in days. From a maintenance perspective, this output represents a planning horizon rather than an exact failure date. The most informative behavior is the evolution of the predicted RUL over time: as more evidence of degradation accumulates, predictions are expected to become more consistent, whereas abrupt shifts in the RUL curve indicate either a change in degradation rate, increased noise, or an operating regime shift. The attention mechanism is particularly useful in this context because it can emphasize time intervals where degradation accelerates, which are often critical for early warning and maintenance scheduling.

8.4 Sensitivity and Uncertainty

RUL estimation is inherently sensitive to both the degradation trajectory and the measurement process. Within this project, the predicted RUL is strongly influenced by the injected fault growth rate, which determines how quickly features and the health representation evolve. Noise injection and feature variability introduce short-term fluctuations that can propagate to RUL updates. The choice of window length and sequence horizon also affects stability: shorter windows improve responsiveness but increase variance, while longer aggregation improves robustness at the cost of latency. In deployment, additional uncertainty arises from model mismatch, since real machines may degrade through multiple mechanisms and under changing operating conditions. For this reason, RUL estimates should be accompanied by confidence bounds or stability indicators when used for operational decisions.

8.5 Role of the Dashboard in Interpretation

The dashboard is not only a visualization tool but also a mechanism for interpreting results in a maintenance-oriented context. Displaying the health indicator, severity progression, and RUL estimate together helps diagnose whether changes in predicted RUL are driven by genuine degradation progression or by short-term variability. Warning messages provide a simple human-readable interface for threshold-based actions, and the controls support repeatable experiments during model tuning. In an industrial setting, such an interface can act as a first layer of decision support, with integration into maintenance management systems as a future extension.

8.6 Limitations and Deployment Considerations

The proposed workflow is validated in a simulation environment and therefore inherits the assumptions and fidelity limitations of the digital twin. Degradation is generated through controlled parameter changes, while real degradation may be non-monotonic and influenced by external factors such as temperature, lubrication, alignment changes, and load variation. In addition, the end-of-life threshold is defined within the simulation context and must be linked to engineering limits or historical failure evidence for real deployment. To reduce the sim-to-real gap, future work should prioritize calibration with measured vibration data, inclusion of operating regime variability, and validation on independent datasets. Despite these limitations, the project demonstrates a complete and extensible framework that combines physics-based simulation with data-driven prognostics and provides a clear pathway toward real-world application.

9 Conclusion and Future Work

This report presented a simulation-driven digital twin framework for a rotating shaft system aimed at predictive maintenance through remaining useful life (RUL) estimation. A physics-based model was implemented in MATLAB Simulink and Simscape and instrumented with virtual sensors to generate vibration, torque, and speed signals. These outputs were processed into condition features and organized into 30-day sequences to support data-driven prognostics. A Bidirectional LSTM model with an attention mechanism was used to predict RUL in days, and the complete workflow was demonstrated through an interactive monitoring dashboard that visualizes degradation severity, health indication, and predicted RUL together with warning messages and operational controls.

The developed system demonstrates how physics-based simulation and machine learning can be combined to build an end-to-end prognostics pipeline when run-to-failure measurements are limited. The dashboard presentation further supports interpretability by connecting degradation progression to predicted remaining life in a form suitable for maintenance planning.

9.1 Future Work

Several extensions can improve the fidelity, generalization, and deployment readiness of the proposed approach. First, the digital twin should be calibrated using measured vibration data from a physical shaft system, including realistic sensor noise and mounting effects. Second, the degradation modeling should be expanded to include multiple mechanisms and operating regimes, such as load changes, speed variation, and combined fault modes, to improve robustness of the learned RUL predictor. Third, uncertainty quantification should be added to provide confidence bounds on RUL estimates, enabling safer decision-making. Finally, the dashboard can be extended with real-time data streaming and automated logging so that the system can operate as a continuous monitoring tool, with potential integration into maintenance management systems.

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